2026-03-23

# Modernising Integer Partitions: Implementing Standard Iterators for the partitions Package

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## Executive Summary

### Overview

The `partitions` package (Robin K. S. Hankin 2006) is a widely-used R utility for enumerating integer partitions and related combinatorial structures. With over 1.3 million downloads, 6000–8000 monthly users, and 36 dependencies, it is critical R infrastructure. However, the package suffers from two key limitations: (1) bulk-generation APIs cause catastrophic memory exhaustion for large problems, and (2) the existing iterator API is non-standard, preventing seamless integration with modern R ecosystems (`foreach`, `itertools`, `parallel`).

This proposal seeks to modernise the `partitions` package by implementing the standard S3 `iterators` protocol (Analytics and Weston 2022), enabling lazy evaluation, efficient memory usage, and interoperability with parallel computing frameworks. We will create a new S3 class `partitions_iter` with `nextElem` methods for the core enumeration functions, along with comprehensive documentation and vignettes demonstrating practical use cases.

### Project Scope

**Primary Deliverable:** An S3 `partitions_iter` class providing scalable, lazy iteration that conforms to the `iterators` interface, enabling direct integration with `foreach()` and `itertools` (Weston and Wickham 2014).

**Secondary Deliverables:**

- Comprehensive Quarto vignette with 5+ worked examples: serial iteration, `foreach()` parallelisation, `itertools` chaining, memory-efficiency comparisons.
- Full test suite covering concurrent iterator behaviour, edge cases, and GitHub issue resolution.
- CRAN-ready code and final review with Robin Hankin.

## Budget & Timeline

We request **$18,331.21** to fund **124 hours** of development. This is a joint costing from Oxford Brookes University and the University of Stirling.

Sam Kamperis (Oxford Brookes) would be allocated 80 hours for the development. Robin Hankin (Stirling) would be allocated 44 hours, including a two-day focused visit in Oxford. There is also a small budget for Robin's travel and accommodation for this visit.

| Item | Cost (USD) |
|---|---|
| Sam Kamperis 80 hours | **$11,042.24** |
| Robin Hankin 44 hours | **$6,783.71** |
| Travel (Hotel + Train to Oxford) | **$505.26** |
| **Total** | **$18,331.21** |

*The travel budget ($505.26) covers Robin Hankin's two-day visit to Oxford for technical collaboration and review.*

## Team

The project is led by the grant applicant (9.7% FTE over 6 months) with **Robin Hankin**, the original `partitions` package author, as a collaborator providing technical guidance, code review, and CRAN liaison.

## Expected Impact

This modernisation unlocks new use cases: practitioners can now leverage the package seamlessly within modern R functional pipelines and parallel frameworks, enabling applications such as high-throughput research tasks in operations research, cryptography, and scientific computing. The new iterator interface allows direct integration with tools like `foreach`, `future`, and `itertools`, supporting reproducible research, teaching, and complex computational workflows that were previously impractical. Previously, workarounds required verbose, error-prone custom code, discouraging adoption in teaching and research, and the lack of standard iterator support prevented the package's use in parallel and distributed computing, limiting its utility in modern data science workflows.

# Signatories

## Project team

**Sam Kamperis** Github
**Robin Hankin** Github

## Contributors

## Consulted

**James Maunder**
**Fridolin Wild** Github
**Inna Skarga-Bandurova** Github

# The Problem

## What Is the Problem?

The `partitions` package (Robin K. S. Hankin 2006) provides two distinct APIs for enumerating partitions:

1. **Bulk-generation functions** (`parts()`, `restrictedparts()`, `diffparts()`): Generate all partitions in memory as a matrix and return them at once.
2. **Stepwise iteration** (`firstpart()` + `nextpart()`, etc.): Users write manual `while` loops, maintaining state vectors and calling low-level stepping functions.

Both approaches have critical limitations.

### Combinatorial Explosion & Memory Crashes

For even moderate values of $n$, the number of partitions grows explosively. The partition count function $p(n)$ grows as:

$$p(n) \sim \frac{1}{4n\sqrt{3}} \exp\left(\pi\sqrt{\frac{2n}{3}}\right)$$

Concretely: $p(50) \approx 204$ million partitions; $p(100) \approx 190$ quadrillion. Users attempting `parts(50)` or larger will exhaust RAM and crash their R session. This is not a minor inconvenience, it is a **hard blocker** for realistic combinatorial research.

### Non-Standard Iterator Protocol

The `firstpart()`/`nextpart()` API does not conform to the R ecosystem's standard iterator interface defined by the `iterators` package (Analytics and Weston 2022). Users cannot:

- Pass partition iterators directly to `foreach()` without custom wrappers.
- Chain partition iterators with `itertools` utilities like `ichain()`, `ifilter()`, or `izip()` (Weston and Wickham 2014).
- Use partition enumeration in standard parallel frameworks designed for iterators.

Instead, users write verbose, error-prone `while` loops:

```
part <- firstpart(20)
results <- list()
while (!islastpart(part)) {
  results[[length(results) + 1]] <- process(part)
  part <- nextpart(part, check=FALSE)
}
```

This pattern is tedious, hard to maintain, and incompatible with lazy evaluation and parallelisation idioms.

## Who Is Affected?

With **1.3 million total downloads** and **6,000–8,000 monthly users**, the `partitions` package serves:

- **Researchers** in combinatorics and number theory performing exhaustive enumeration.
- **Statisticians** using partitions for resampling and bootstrap methods.
- **Operations researchers** solving integer partition problems.
- **Parallel computing practitioners** seeking to distribute combinatorial tasks across clusters.

## Why Is This a Problem?

1. **Ecosystem Misalignment:** Modern R functional programming (via `foreach`, `itertools`, `furrr`) assumes standard iterator interfaces. `partitions` is an exception, creating friction and requiring custom adapters.
2. **Memory Degradation:** Researchers working with realistic problem sizes cannot use convenient bulk-generation APIs and must resort to manual state management, introducing bugs and reduced code clarity.
3. **Concurrency Barriers:** Parallel frameworks cannot transparently distribute work over partition enumerations because the non-standard API provides no standard way to yield values on demand.

## What Will Solving This Enable?

Implementing standard iterators will:

- **Enable lazy evaluation:** Users iterate over arbitrarily large partition streams without materialising all values in memory.
- **Unlock parallel workflows:** Standard iterators integrate seamlessly with `foreach()`, `parallel::makeCluster()`, and `furrr`.
- **Improve interoperability:** Partition iterators can be composed with other `itertools` operations (filtering, zipping, chaining).
- **Reduce cognitive load:** Users write familiar, standard R code instead of custom loops.

## Existing Work & Precedent

Standard iterator support is well-established throughout the R ecosystem:

- The `iterators` package (Analytics and Weston 2022) defines the S3 `iter` class and `nextElem` generic method.
- The `itertools` package (Weston and Wickham 2014) provides composable combinators (`chain`, `filter`, `zip`) that work with any standard iterator.
- Packages like `foreach` and `furrr` assume standard iterator interfaces for parallel iteration.

The `partitions` package simply needs to align with these established conventions. The existing `firstpart()`/`nextpart()` API will remain for backwards compatibility; we will **add** (not replace) the standard interface.

# The Proposal

## Overview

We will modernise the `partitions` package by implementing the standard S3 `iterators` protocol (Analytics and Weston 2022), enabling lazy, memory-efficient enumeration of partitions with seamless integration into the modern R ecosystem. The core deliverable is a new `partitions_iter` S3 class providing the `nextElem` method, allowing users to pass partition enumerations directly into `foreach()` loops, `itertools` combinators, and parallel frameworks.

The solution preserves all existing APIs for backwards compatibility while adding a thin, well-tested layer that conforms to established iterator conventions. This approach minimises risk: we leverage the battle-tested `firstpart()`/`nextpart()` enumeration logic rather than replacing it, and we add comprehensive documentation and examples to guide users.

## Detail

### Minimum Viable Product

A working MVP consists of:

1. **S3 class `partitions_iter`** with fields `state` (current partition vector), `n` (the target integer), and `type` (enumeration type: "unrestricted", "restricted", "distinct", etc.).
2. **Method `nextElem.partitions_iter()`** that calls the appropriate stepping function (`nextpart()`, `nextdiffpart()`, etc.), returns the current state, advances state via the `<<-` operator, and throws `'StopIteration'` when exhausted.
3. **Factory functions** `iparts()`, `irestrictedparts()`, `idiffparts()` that return an iterator (S3 object) for lazy enumeration.
4. **Basic `testthat` tests** verifying that iterators produce the same results as bulk-generation functions, and that multiple iterators coexist without state collision.

Example usage:

```r
library(partitions)
library(foreach)

# Serial enumeration
it <- iparts(15)
while (!islastpart(it$state)) {
  print(nextElem(it))
}

# Parallel iteration with foreach
results <- foreach(part = iparts(20), .combine = c) %dopar% {
  sum(part)  # Example: sum each partition
}
```

### Architecture

**R-level:** - `iparts(n)` → creates an S3 object of class `c("partitions_iter", "abstractiter", "iter")` - Object stores `state` (current partition vector), `n`, and `type` - `nextElem.partitions_iter()` closure uses `<<-` to mutate state in the enclosing environment

**C-level:** - No changes required; existing `c_nextpart()`, `c_nextrestrictedpart()`, etc. are reused as-is - These functions already provide re-entrant state mutation (each partition vector is independent)

**Integration:** - The `iter()` generic method (inherited from `iterators` package) returns the iterator itself - `foreach()` and `itertools` function detect the `iter` class and call `nextElem()` in a loop - State isolation is automatic: each call to `iparts()` creates a new closure with its own `state` variable

### Assumptions

1. **The C-level enumeration logic is correct and mature.** We are not reimplementing it; only wrapping it.
2. **The `iterators` package API is stable.** (It is; the package is mature and widely used.)
3. **Users have installed `iterators` as a dependency.** (We will add it to `DESCRIPTION` with `Imports: iterators`.)
4. **Multiple iterators can coexist without state collision.** (They can, because each iterator closure has its own environment.)

### Failure Modes & Mitigation

| Failure Mode | Likelihood | Mitigation |
|---|---|---|
| Iterator state mutations are slow due to copy-on-modify semantics | Low | Partition vectors are relatively small ($< 1000$ elements typically); overhead is negligible. |
| Users expect `stop('StopIteration')` instead of `error()` | Low | `nextElem()` from `iterators` package already standardises this; we follow the convention. |
| Parallel workers cannot access iterator closures | Low | Each worker receives a serialised copy of the iterator object (state vector, n, type); no shared mutable state. |
| Restrictedparts edge cases fail (e.g., `m > n`) | Medium | Comprehensive unit tests; review existing test coverage; coordinate with Robin for edge-case validation. |

### External Dependencies

- **`iterators`** (CRAN): S3 class definitions and `nextElem` generic. Must be added to `Imports:` in `DESCRIPTION`.
- **`foreach`** (CRAN, optional): User convenience for parallel enumeration. Not a hard dependency.
- **`itertools`** (CRAN, optional): User convenience for composing iterators. Not a hard dependency.
- **`testthat`** (CRAN, existing dev dependency): Test infrastructure.

All dependencies are mature, actively maintained, and widely adopted. No risk of ecosystem instability.

### Evidence of Impact

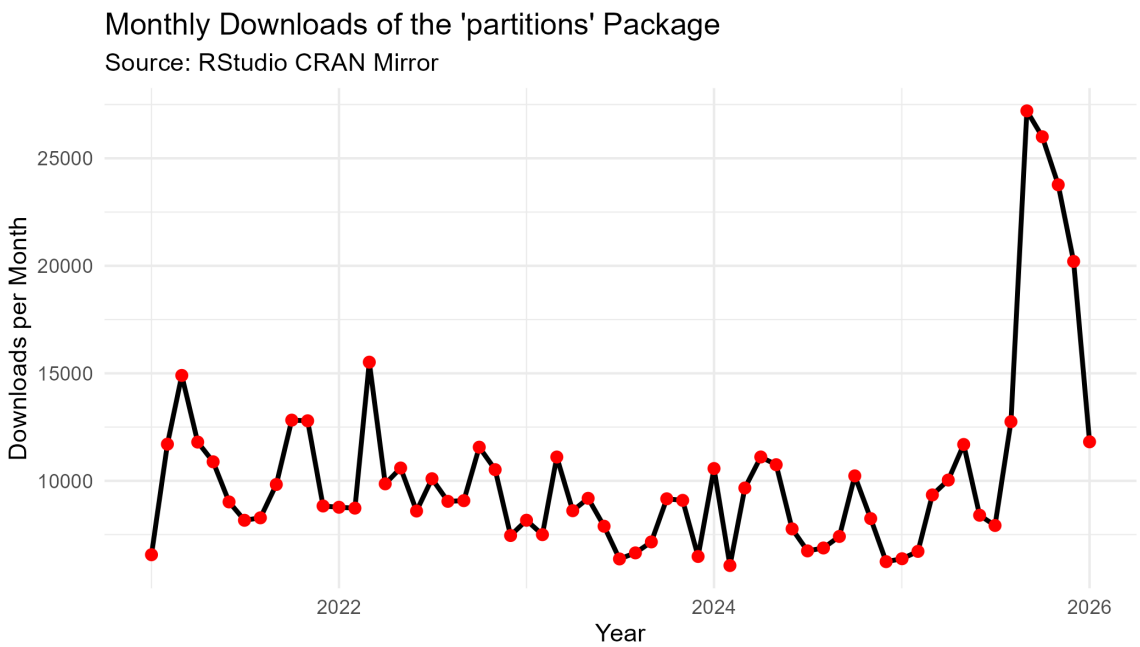


**Figure 1:** Monthly Downloads of the partitions package

The plot above demonstrates the `partitions` package's significant and sustained user adoption. With ~**1.3 million cumulative downloads** and **6,000–8,000 monthly active users**, the package is a foundational tool in the R ecosystem. This level of usage—comparable to well-established data manipulation and statistical packages—justifies investment in modernising its infrastructure to integrate with contemporary R practices (lazy evaluation, functional programming, parallelisation).

Users across combinatorics research, statistics, operations research, and scientific computing depend on this package. Enabling standard iterator support will unlock new use cases and remove artificial barriers to productivity for researchers working with large combinatorial spaces.

## Project Plan

### Phase 0: Start-up (June 2026, Weeks 1–4)

- Establish collaboration infrastructure: GitHub issue tracking, pull request templates, contributor guidelines.
- Review existing `partitions` codebase: C API, R wrapper functions, test coverage, and GitHub open issues.
- Draft S3 `partitions_iter` class structure and method signatures.

- Set up continuous integration (CI) via GitHub Actions to ensure all changes pass `R CMD check` and `testthat` suite.
- Finalise scope and technical design with Robin Hankin.

**Effort:** 8 hours (Sam Kamperis), 8 hours (Robin Hankin)

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## Phase 1: Core Implementation (July 2026, Weeks 5–8)

- Implement `iparts()`, `irestrictedparts()`, `idiffparts()` factory functions.
- Implement `nextElem.partitions_iter()` method for each enumeration type.
- Ensure backward compatibility: existing `parts()`, `firstpart()`, `nextpart()` APIs remain unchanged.
- Write unit tests (`testthat`) covering basic iteration, edge cases (n=1, n=0, m>n, etc.), and concurrent iterator coexistence.
- Robin Hankin: technical guidance and review (4 hours).

**Effort:** 16 hours (Sam Kamperis), 8 hours (Robin Hankin)

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## Phase 2: Integration & Testing (August 2026, Weeks 9–12)

- Integrate with `foreach` package: test that `foreach(part = iparts(n)) %do%` and `%dopar%` work correctly.
- Integrate with `itertools` package: verify chaining via `itertools::ichain()`, filtering via `ifilter()`, etc.
- Address outstanding GitHub issues (bug fixes, documentation improvements) in the context of the iterator framework.
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## Phase 3: Vignette & Documentation (September 2026, Weeks 13–18)

- Write a detailed Quarto vignette (5+ worked examples):
  1. Serial enumeration: Basic `while` loop with `nextElem()`.
  2. `foreach()` parallelisation: Distributing partition enumeration across cores/cluster.
  3. `itertools` composition: Chaining, filtering, zipping partition iterators.
  4. Memory-efficiency comparison: Bulk generation (`parts(n)`) vs. lazy iteration (`iparts(n)`) for large n.
  5. Real-world use case: Sampling partitions, computing partition statistics, parallel optimisation.
- Update package `README.md` with modern iterator examples.

- Ensure all R documentation (`.Rd` files) reflects new iterator functions.
- Add CRAN-compliant licensing headers if necessary.
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## Phase 4: Review & Release (October 2026, Weeks 19–22)

- Code review with Robin Hankin; incorporate feedback.
- Final `R CMD check` and comprehensive testing.
- Prepare CRAN submission: ensure all checks pass, update `NEWS.md`, verify compatibility.
- Integrate 5-hour contingency for extended review, issue triage, or unexpected fixes.
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## Phase 5: Publicity & Dissemination (November 2026, Weeks 23–26)

- Submit to CRAN.
- Address CRAN feedback (if any) and release new package version.
- Announce release on R Consortium blog and community channels.
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## Budget & Funding Plan

**Total Funding:** $18,331.21 **Duration:** 6 months (1 June – 30 November 2026)
**Total Hours:** 124 (Sam Kamperis: 80, Robin Hankin: 44)

| Phase | Dates | Sam Kamperis (hrs) | Robin Hankin (hrs) | Deliverables | Funding (%) |
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| **Phase 0: Start-up** | June | 8 | 8 | Collaboration infra, design doc, CI setup | 12% |
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| **Total** | June–Nov | **80** | **44** | Full modernisation + CRAN release | **100%** |

## Technical Delivery

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- Submit to CRAN.
- Address CRAN feedback (if any) and release new package version.
- Announce release on R Consortium blog and community channels.
- Robin Hankin: technical guidance and review (4 hours).

**Effort:** 8 hours (Sam Kamperis), 4 hours (Robin Hankin)

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## Other Aspects

### Availability & Licensing

The `partitions` package is licensed under GPL-3. All code, tests, and documentation produced by this project will be released under the same licence, ensuring consistency and compatibility.

Code will be hosted on the official GitHub repository (github.com/RobinHankin/partitions) with pull requests and issues tracked transparently.

### Publicity & Dissemination

- **ISC Blog Post (Mid-project, ~July 2026):** Summary of modernisation goals and progress.
- **Delivery Blog Post (Post-release, ~December 2026):** Technical introduction to the new iterator API, with code examples and benchmarks.
- **R Community Channels:** Announcements on R Weekly, R Discourse, and relevant social media.
- **Conference Consideration:** Abstract submission to useR! 2027 if timeline permits.

## Budget & Funding Plan

### Budget Allocation

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| **Total** | June–Nov | **80** | **44** | Full modernisation + CRAN release | **100%** |

**Payment Schedule:** 50% of the grant is paid upon contract signing, and 50% upon completion. The contingency (5 hours) is included in Phase 4 for extended review or unexpected issues.

# Success

## Definition of Done

Success is measured by the following concrete, verifiable outcomes:

1. **Code merged and released:** All iterator implementation code is merged into the main branch of the GitHub repository and accepted by Robin Hankin for inclusion in a new CRAN release (version number TBD, e.g., `partitions >= 1.11-0`).

2. **Standard iterator conformance:** The new `partitions_iter` class is fully compliant with the `iterators` package S3 interface. Users can pass iterators directly to `foreach()` and `itertools` functions without custom wrappers.

3. **Comprehensive test coverage:** A full `testthat` suite validates:

   - Correctness: iterators produce identical results to bulk-generation functions.

- Concurrency: multiple iterators coexist and advance independently.
- Edge cases: n=0, n=1, m>n, restrictedparts with various constraints.
- Interoperability: `foreach()`, `itertools::ichain()`, `itertools::ifilter()`, and parallel backends.

4. **Published vignette:** A detailed Quarto vignette is included in the package, demonstrating 5+ worked examples spanning serial iteration, parallelisation, composition, and real-world use cases.

5. **GitHub issues resolved:** Outstanding GitHub issues relevant to iteration are addressed and closed (or documented as out-of-scope with clear justification).

## Measuring Success

We will track the following metrics throughout the project:

| Metric | Success Criterion | Measurement Method |
|---|---|---|
| **Code Coverage** | 90% of new iterator code covered by tests | `covr` package report |
| **CRAN Compliance** | All `devtools::check()` and `R CMD check` tests pass with zero errors/warnings | CI pipeline; manual CRAN dry-run |
| **Interoperability** | `foreach()` and `itertools` examples execute without error | Unit tests + manual vignette examples |
| **Memory Efficiency** | Lazy iteration uses <10% memory vs. bulk generation for large n | Benchmark script comparing `parts(50)` vs. `iparts(50)` |
| **Documentation Quality** | Vignette is comprehensible to users unfamiliar with iterators | User testing (informal review by collaborators) |
| **GitHub Issues** | 80% of open iterator-related issues resolved or documented | GitHub issue tracker review |
| **CRAN Release** | New version submitted and accepted by CRAN | Official email confirmation from CRAN maintainers |

### Progress Checkpoints

- **End of June (Milestone 1):** Design doc approved by Robin; CI infrastructure operational; initial test suite in place.
- **End of August (Milestone 2–3):** `iparts()`, `irestrictedparts()`, `idiffparts()` factory functions implemented and tested; `foreach()` integration working.
- **End of September (Milestone 3):** `itertools` integration functional; GitHub issues triaged and addressed.
- **End of October (Milestone 4):** Vignette drafted and reviewed; all documentation updated.
- **End of November (Milestone 5):** CRAN submission completed; public announcement released.

## Future Work

### Short-term (Post-Release)

- **Additional partition types:** Extend iterator support to `setparts()`, `blockparts()`, `compositions()` if demand warrants.
- **Performance optimisation:** Profile and optimise the `nextElem()` closure if memory/CPU overhead becomes significant in high-throughput scenarios.
- **Educational resources:** Publish a blog post or tutorial on the `partitions` package ecosystem for the R community blog.

### Medium-term (6–12 Months)

- **Parallel sampling:** Develop advanced itertools, e.g., `isample_partitions()` for unbiased random partition sampling in parallel.
- **Integration with `furrr`:** Ensure seamless compatibility with `furrr::future_map()` and other functional programming packages.
- **Package ecosystem integration:** Collaborate with authors of related packages (e.g., combinatorics, optimisation) to showcase shared iterator workflows.

### Long-term (1–2 Years)

- **Persistent state storage:** Support serialisation of iterator state to disk, enabling resumption of long-running enumerations across sessions.
- **Distributed computing:** Explore Apache Spark / Hadoop integration for massive-scale partition enumeration in clusters beyond standard R parallelism.
- **Combinatorics library expansion:** Consider generalising the iterator pattern to other combinatorial enumerations (permutations, combinations, Young tableaux, etc.) as a separate infrastructure project.

Analytics, Revolution, and Steve Weston. 2022. *Iterators: Provides Iterator Construct*. https://doi.org/10.32614/CRAN.package.iterators.

Robin K. S. Hankin. 2006. "Additive integer partitions in R." *Journal of Statistical Software, Code Snippets* 16 (May).

Weston, Steve, and Hadley Wickham. 2014. *Itertools: Iterator Tools*. https://doi.org/10.32614/CRAN.package.itertools.